Pleiotropy

This exercise was designed to give practical experience identifying cross phenotype associations using both univariate and multivariate methods and then dissecting these cross phenotype associations to determine if they show evidence of biological and/or mediated pleiotropy.

A population-based dataset with 3000 subjects and two quantitative traits (Trait 1 and Trait 2) along with 2000 SNPs on chromosome 1 were simulated. Let's assume that Trait 1 was measured 20 years prior to Trait 2 (i.e. Trait 1 will act as the mediator in our mediation analysis). The two quantitative traits are correlated and there are markers associated with one or both phenotypes as well as unassociated as well as not being associated.

Adapted from an exercise by Andrew DeWan, PhD, MPH.

Data overview

The data is in PLINK binary-file format under `data/`: `pleiotropy_exercise.bed`, `pleiotropy_exercise.bim`, and `pleiotropy_exercise.fam`. `data/pleiotropy_exercise_phenotypes.txt` is a ready-made phenotype file. The dataset has already been QC-ed.

How this notebook is organized

1. **Univariate analysis** — test each SNP against Trait 1 and Trait 2 separately, visualize overlap, and LD-clump the genome-wide significant hits.
2. **Multivariate analysis** — jointly test both traits to find additional suggestive/significant regions the univariate scans might have missed.
3. **Mediation analysis** — for the SNPs identified above, test whether their effect on Trait 2 runs *through* Trait 1 (mediated pleiotropy) or acts independently on both traits (biological pleiotropy).

Summary table

A summary table is included at the end of the exercise that you will want to fill out as you are working through this exercise. This will help keep track of the SNPs you select for the mediation analysis as well as the interpretation of the results at the end of the exercise.

Program documentation

- **PLINK**: 1.90 docs
- **R**: r-project.org — from within R, `help()` documents any command

References

- Chang CC, Chow CC, Tellier LCAM, Vattikuti S, Purcell SM, Lee JJ (2015) Second-generation PLINK: rising to the challenge of larger and richer datasets. GigaScience, 4.
- Purcell S, Neale B, Todd-Brown K, Thomas L, Ferreira MAR, Bender D, Maller J, Sklar P, de Bakker PIW, Daly MJ & Sham PC (2007) PLINK: a toolset for whole-genome association and population-based linkage analysis. *American Journal of Human Genetics*, 81:559-575.
- R Core Team (2025), R: A Language and Environment for Statistical Computing, R Foundation for Statistical Computing
- Tingley D, Yamamoto T, Hirose K, Keele L, Imai K (2014). "mediation: R Package for Causal Mediation Analysis." Journal of Statistical Software, 59(5), 1–38. https://www.jstatsoft.org/v59/i05/.

> **Kernel:** run the code cells in this exercise with the **Bash** kernel.

1. Univariate Analysis

Conduct a univariate analysis (using --linear) in PLINK for both datasets and both traits

Note: You will need to use the --pheno/--pheno-name commands to specify the phenotype file and phenotype name.

Run the analysis using PLINK

In [1]:
```
# Run linear association analysis for Trait 1, including sex as a covariate
plink \
 --bfile ./data/pleiotropy_exercise \
 --pheno ./data/pleiotropy_exercise_phenotypes.txt \
 --pheno-name Trait1 \
 --sex \
 --linear \
 --out ./output/Trait1

grep 'TEST' ./output/Trait1.assoc.linear > ./output/Trait1_snp.assoc.linear
grep 'ADD' ./output/Trait1.assoc.linear >> ./output/Trait1_snp.assoc.linear
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
9/
(C) 2005-2024 Shaun Purcell, Christopher Chang   GNU General Public License
v3
Logging to ./output/Trait1.log.
Options in effect:
  --bfile ./data/pleiotropy_exercise
  --linear
  --out ./output/Trait1
  --pheno ./data/pleiotropy_exercise_phenotypes.txt
  --pheno-name Trait1
  --sex

Note: --sex flag deprecated.  Use e.g. "--linear sex".
7836 MB RAM detected; reserving 3918 MB for main workspace.
2000 variants loaded from .bim file.
30000 people (14915 males, 15085 females) loaded from .fam.
30000 phenotype values present after --pheno.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 30000 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Total genotyping rate is exactly 1.
2000 variants and 30000 people pass filters and QC.
Phenotype data is quantitative.
Writing linear model association results to ./output/Trait1.assoc.linear ...
done.
```

In [2]:

```
# Repeat for Trait 2
plink \
 --bfile ./data/pleiotropy_exercise \
 --pheno ./data/pleiotropy_exercise_phenotypes.txt \
 --pheno-name Trait2 \
 --sex \
 --linear \
 --out ./output/Trait2

grep 'TEST' ./output/Trait2.assoc.linear > ./output/Trait2_snp.assoc.linear
grep 'ADD' ./output/Trait2.assoc.linear >> ./output/Trait2_snp.assoc.linear
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)            cog-genomics.org/plink/1.
9/
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v3
Logging to ./output/Trait2.log.
Options in effect:
  --bfile ./data/pleiotropy_exercise
  --linear
  --out ./output/Trait2
  --pheno ./data/pleiotropy_exercise_phenotypes.txt
  --pheno-name Trait2
  --sex

Note: --sex flag deprecated.  Use e.g. "--linear sex".
7836 MB RAM detected; reserving 3918 MB for main workspace.
2000 variants loaded from .bim file.
30000 people (14915 males, 15085 females) loaded from .fam.
30000 phenotype values present after --pheno.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 30000 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Total genotyping rate is exactly 1.
2000 variants and 30000 people pass filters and QC.
Phenotype data is quantitative.
Writing linear model association results to ./output/Trait2.assoc.linear ...
done.
```

Visualize data and identify overlapping signals

Try visualizing the data by creating a Hudson plot and identify genome-wide significant SNPs (p-value ≤ 5×10^{-8}) that overlap for both traits using R. This will give you some sense of the overlapping signals between the two association analyses.

This will be ran using an external R script `scripts/generate_plots_overlap_signals_univariate.R`, the code used to generate the plot and identify significant SNPs is provided below:

```r
library(hudson)

dat1<-read.table("./output/Trait1_snp.assoc.linear",header=T)
dat2<-read.table("./output/Trait2_snp.assoc.linear",header=T)
names(dat1)<-c("CHR", "SNP", "POS", "A1", "TEST", "NMISS", "BETA",
"STAT", "pvalue")
names(dat2)<-(names(dat1))
gmirror(top=dat1, bottom=dat2, tline=5e-08, bline=5e-08,
    toptitle="Trait1", bottomtitle = "Trait2",
    highlight_p = c(0.00000005,0.00000005), highlighter="green",
    file = './output/pleiotropy_hudson', res = 300, type = 'pdf')

Trait1 <- read.table("./output/Trait1_snp.assoc.linear", header =
T)
Trait2 <- read.table("./output/Trait2_snp.assoc.linear", header =
```

```
T)
SigTrait1 <- subset(Trait1, P<0.00000005)
SigTrait2 <- subset(Trait2, P<0.00000005)
intersect(SigTrait1$SNP, SigTrait2$SNP)
```

In [3]:
```
# Generate the Hudson plot and print genome-wide significant SNPs shared bet
Rscript - <<'EOF'
pdf(file = NULL)

library(hudson)

dat1<-read.table("./output/Trait1_snp.assoc.linear",header=T)
dat2<-read.table("./output/Trait2_snp.assoc.linear",header=T)
names(dat1)<-c("CHR", "SNP", "POS", "A1", "TEST", "NMISS", "BETA", "STAT", "
names(dat2)<-(names(dat1))
gmirror(top=dat1, bottom=dat2, tline=5e-08, bline=5e-08,
    toptitle="Trait1", bottomtitle = "Trait2",
    highlight_p = c(0.00000005,0.00000005), highlighter="green",
    file = './output/pleiotropy_hudson', res = 300, type = 'pdf')
graphics.off()

Trait1 <- read.table("./output/Trait1_snp.assoc.linear", header = T)
Trait2 <- read.table("./output/Trait2_snp.assoc.linear", header = T)
SigTrait1 <- subset(Trait1, P<0.00000005)
SigTrait2 <- subset(Trait2, P<0.00000005)
intersect(SigTrait1$SNP, SigTrait2$SNP)
EOF
```

```
Loading required package: ggplot2
Loading required package: gridExtra
Loading required package: colorspace
Loading required package: ggiraph
Loading required package: cowplot
[1] "Saving plot to ./output/pleiotropy_hudson.pdf"
TableGrob (2 x 1) "arrange": 2 grobs
  z     cells    name           grob
1 1 (1-1,1-1) arrange gtable[arrange]
2 2 (2-2,1-1) arrange gtable[arrange]
Warning messages:
1: No shared levels found between `names(values)` of the manual scale and th
e
data's colour values.
2: No shared levels found between `names(values)` of the manual scale and th
e
data's colour values.
[1] "rs138"  "rs139"  "rs140"  "rs141"  "rs296"  "rs299"  "rs1138" "rs1448"
```

LD clumping

As you can see, there are some genome-wide significant SNPs that are adjacent or close to each other. To explore whether or not these are independent associations, let's perform some simple LD clumping. You will want to retain through the index SNP identified for each clumped region. You will also want to retain through any SNPs from above that were not part of a clumped region.

In [4]:
```
# LD clump the genome-wide significant hits from both traits
plink \
 --bfile ./data/pleiotropy_exercise \
 --clump ./output/Trait1_snp.assoc.linear,./output/Trait2_snp.assoc.linear \
 --clump-kb 250 \
 --clump-p1 5e-8 \
 --clump-p2 5e-8 \
 --clump-r2 0.2 \
 --clump-replicate \
 --clump-verbose \
 --out ./output/Trait1_Trait2_clump
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
9/
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v3
Logging to ./output/Trait1_Trait2_clump.log.
Options in effect:
  --bfile ./data/pleiotropy_exercise
  --clump ./output/Trait1_snp.assoc.linear,./output/Trait2_snp.assoc.linear
  --clump-kb 250
  --clump-p1 5e-8
  --clump-p2 5e-8
  --clump-r2 0.2
  --clump-replicate
  --clump-verbose
  --out ./output/Trait1_Trait2_clump

7836 MB RAM detected; reserving 3918 MB for main workspace.
2000 variants loaded from .bim file.
30000 people (14915 males, 15085 females) loaded from .fam.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 30000 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Total genotyping rate is exactly 1.
2000 variants and 30000 people pass filters and QC.
Note: No phenotypes present.
--clump: 2 clumps formed from 143 top variants.
Results written to ./output/Trait1_Trait2_clump.clumped .
```

2. Multivariate Analysis

In this section we will use multivariate analysis to see if there are additional SNPs/regions to explore, prior to moving on to dissecting the cross phenotype associations. We will only consider additional regions that are genome-wide suggestive (p-value $\leq 5 \times 10^{-6}$) for both phenotypes.

Run the analysis on Traits 1 & 2

We use the `MultiPhen` R package (`mPhen`), which jointly models Trait1 and Trait2 as predictors of each SNP (ordinal regression) and reports a single joint p-value per SNP.

Note: this tests all 2000 SNPs and takes a couple of minutes to run.

In [5]:
```
Rscript - <<'EOF'
suppressMessages(library(MultiPhen))

# Read genotype and phenotype data
bim <- read.table("./data/pleiotropy_exercise.bim",
                  col.names = c("chr", "snp", "cm", "pos", "a1", "a2"))
key <- paste(bim$chr, bim$pos, sep = "_")

geno  <- mPhen.readGenotypes("./data/pleiotropy_exercise", opts = mPhen.opti
colnames(geno$genoData) <- bim$snp[match(colnames(geno$genoData), key)]
pheno <- mPhen.readPhenoFiles("./data/pleiotropy_exercise_phenotypes.txt",
                              opts = mPhen.options("pheno.input"))

opts <- mPhen.options(c("regression", "pheno.input"))
opts$mPhen.scoreTest <- TRUE

# Run the joint association test across all SNPs in batches of 100
n <- ncol(geno$genoData)
batch <- 100
chunks <- list()
for (start in seq(1, n, by = batch)) {
    end <- min(start + batch - 1, n)
    r <- mPhen(geno$genoData[, start:end, drop = FALSE], pheno$pheno,
               phenotypes = c("Trait1", "Trait2"), opts = opts)
    chunks[[length(chunks) + 1]] <- data.frame(
        SNP         = dimnames(r$Results)[[2]],
        P           = as.numeric(r$Results["all", , "JointModel", "pvalue"])
        Beta_Trait1 = as.numeric(r$Results["all", , "Trait1", "beta"]),
        Beta_Trait2 = as.numeric(r$Results["all", , "Trait2", "beta"])
    )
}
out <- do.call(rbind, chunks)

write.table(out, "./output/Trait1_Trait2_multiphen.txt",
            sep = "\t", row.names = FALSE, quote = FALSE)
cat("Wrote", nrow(out), "SNPs to ./output/Trait1_Trait2_multiphen.txt\n")
cat("Genome-wide significant (P<5e-8):", sum(out$P < 5e-8), "\n")
EOF
```

```
[1] 30000  2000
[1] "excluding 0 samples based on exclusion criteria"
[1] "excluding 0 samples based on exclusion criteria"
[1] "excluding 0 samples based on exclusion criteria"
[1] "excluding 0 samples based on exclusion criteria"
[1] "excluding 0 samples based on exclusion criteria"
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[1] "excluding 0 samples based on exclusion criteria"
[1] "excluding 0 samples based on exclusion criteria"
[1] "excluding 0 samples based on exclusion criteria"
[1] "excluding 0 samples based on exclusion criteria"
Wrote 2000 SNPs to ./output/Trait1_Trait2_multiphen.txt
Genome-wide significant (P<5e-8): 153
```

Identify multivariate signals

Identify the intersection of SNPs that are genome-wide significant in the multivariate analysis and at least suggestive for each trait in the univariate analysis, i.e. we want to make sure that both traits are contributing to the multivariate signal.

Select the additional SNPs that are identified from the intersection of the multivariate analysis and genome-wide suggestive lists for both traits that were not in your original list.

`./output/Trait1_Trait2_multiphen.txt` has one row per SNP: `P` is the joint model's p-value (this is the "MV (P)" column in the summary table), and `Beta_Trait1` / `Beta_Trait2` are each trait's contribution to that joint model — use these for the "MV Beta" columns.

In [6]:
```
# Extract SNP IDs below a p-value threshold from a results file
filter_snps() {
    awk -v thresh="$2" '
        NR == 1 {
            for (i = 1; i <= NF; i++) {
                if ($i == "SNP") s = i
                if ($i == "P")   p = i
            }
            if (!s || !p) { print "missing SNP or P column in " FILENAME > "
            next
        }
        $p + 0 == $p && $p + 0 < thresh { print $s }
```

```
    ' "$1" | sort -u
}

# Suggestive threshold for each univariate trait; genome-wide significance f
filter_snps ./output/Trait1_snp.assoc.linear       0.000005   > ./output/sug
filter_snps ./output/Trait2_snp.assoc.linear       0.000005   > ./output/sug
filter_snps ./output/Trait1_Trait2_multiphen.txt   0.00000005 > ./output/sig

# Keep only SNPs that appear in all three lists
comm -12 ./output/sugg_trait1.snps ./output/sugg_trait2.snps \
| comm -12 - ./output/sig_multi.snps \
| tee ./output/shared.snps
```

```
rs1138
rs1166
rs125
rs135
rs1361
rs137
rs138
rs139
rs140
rs141
rs1448
rs295
rs296
rs298
rs299
rs300
rs920
rs921
rs923
```

LD clumping

Re-run the LD clumping with a suggestive threshold to see if these additional SNPs clump with your existing clumps or are new potential regions to explore.

In [7]:

```
# Re-run LD clumping at the suggestive threshold (p < 5e-6)
plink \
 --bfile ./data/pleiotropy_exercise \
 --clump ./output/Trait1_snp.assoc.linear,./output/Trait2_snp.assoc.linear \
 --clump-p1 0.000005 \
 --clump-p2 0.000005 \
 --clump-r2 0.2 \
 --clump-replicate \
 --clump-verbose \
 --out ./output/Trait1_Trait2_clump_suggestive
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
9/
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v3
Logging to ./output/Trait1_Trait2_clump_suggestive.log.
Options in effect:
  --bfile ./data/pleiotropy_exercise
  --clump ./output/Trait1_snp.assoc.linear,./output/Trait2_snp.assoc.linear
  --clump-p1 0.000005
  --clump-p2 0.000005
  --clump-r2 0.2
  --clump-replicate
  --clump-verbose
  --out ./output/Trait1_Trait2_clump_suggestive

7836 MB RAM detected; reserving 3918 MB for main workspace.
2000 variants loaded from .bim file.
30000 people (14915 males, 15085 females) loaded from .fam.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 30000 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Total genotyping rate is exactly 1.
2000 variants and 30000 people pass filters and QC.
Note: No phenotypes present.
--clump: 6 clumps formed from 205 top variants.
Results written to ./output/Trait1_Trait2_clump_suggestive.clumped .
```

3. Mediation analysis

Now that you've assembled a candidate list of cross-phenotype SNPs (genome-wide significant in both traits, LD-clump index SNPs, and any additional multivariate hits), this section tests each one for **biological vs. mediated pleiotropy** — see the explanation in the introduction above. For each SNP, we extract its genotype, then fit the two regression models. The `mediation` package needs (SNP → Trait1, and Trait1 + SNP → Trait2) and let it decompose the SNP's total effect on Trait2 into the ACME (mediated through Trait1) and ADE (direct, not through Trait1) components.

Extract data for each SNP

For each SNP that you have identified as a cross phenotype association, you will need to extract this data from the original plink files and create a genotype file that is coded as `0|1|2` for the genotypes. This can be done in PLINK using the `--recodeA` command and the `--extract` command by providing a file with the list of SNPs. This will give you a `.raw` genotype file with only the SNPs that you will be using in the mediation analysis.

Indicate which single SNP you want to run the mediation analysis with — `--extract` needs a plain text file with one SNP ID per line. List the SNPs you selected above: the genome-wide significant overlaps from Step 1, each LD-clump index SNP, and any additional SNPs from the multivariate step (`./output/shared.snps`). The cell below creates a starter file with the worked example this notebook's mediation step uses (`rs125`) — replace its contents with your own selected SNPs before moving on.

In [8]:
```
# List the SNPs to include in the mediation analysis — one per line
# Replace rs125 with your selected SNPs before running
cat > ./output/mediation_snps.txt <<'EOF'
rs125
EOF
```

In [9]:
```
# Extract genotypes for the selected SNPs, coded as 0/1/2 (additive)
plink \
 --bfile ./data/pleiotropy_exercise \
 --recodeA \
 --extract ./output/mediation_snps.txt \
 --out ./output/snps_for_mediation
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
9/
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Note: --recodeA flag deprecated.  Use "--recode A ...".
Logging to ./output/snps_for_mediation.log.
Options in effect:
  --bfile ./data/pleiotropy_exercise
  --extract ./output/mediation_snps.txt
  --out ./output/snps_for_mediation
  --recode A

7836 MB RAM detected; reserving 3918 MB for main workspace.
2000 variants loaded from .bim file.
30000 people (14915 males, 15085 females) loaded from .fam.
--extract: 1 variant remaining.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 30000 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Total genotyping rate is exactly 1.
1 variant and 30000 people pass filters and QC.
Note: No phenotypes present.
--recode A to ./output/snps_for_mediation.raw ... done.
```

Run mediation analysis

Conduct a mediation analysis in R using the mediation R library. A sample code for 1000 simulations is provided below. In real life application, you will need to replace `SNP` with the variable name for the SNP you are investigating and repeat the code for each SNP that you have selected.

Sample code:

```
library(mediation)
genotypes <- read.table("./output/snps_for_mediation.raw",
header=T)
phenotypes <-
read.table("./data/pleiotropy_exercise_phenotypes.txt", header=T)
combined <- merge(genotypes,phenotypes)
med.fit <- lm(Trait1~SNP, data=combined)
out.fit <- lm(Trait2~Trait1+SNP, data=combined)
med.out <- mediate(med.fit,out.fit,treat="SNP", mediator="Trait1",
boot=TRUE, boot.ci.type="bca", sims=1000)
summary(med.out)
```

Note: The more simulations (sims) you specify in the med.out step the more accurate the CI and p-value estimates will be, however, this can also be time-consuming. If this step is taking a substantial amount of time (>20 minutes) you may want to reduce the number of simulations for the purposes of completing the exercise.

For this minimal working example we run 50 simulations. Although `mediation` is an R package, don't switch kernels — R is invoked with `Rscript` so the notebook stays on the bash kernel throughout.

In [10]:
```
Rscript - <<'EOF'
library(mediation)

genotypes  <- read.table("./output/snps_for_mediation.raw", header=T)
phenotypes <- read.table("./data/pleiotropy_exercise_phenotypes.txt", header
combined   <- merge(genotypes, phenotypes)

# Model the SNP's effect on Trait1 (mediator) and Trait2 (outcome)
med.fit <- lm(Trait1 ~ rs125_0, data=combined)
out.fit <- lm(Trait2 ~ Trait1 + rs125_0, data=combined)

# Estimate direct and indirect effects using 50 bootstrap simulations
set.seed(42)
med.out <- mediate(med.fit, out.fit, treat="rs125_0", mediator="Trait1",
                   boot=TRUE, boot.ci.type="bca", sims=50)
summary(med.out)
EOF
```

```
Loading required package: MASS
Loading required package: Matrix
Loading required package: mvtnorm
Loading required package: sandwich
mediation: Causal Mediation Analysis
Version: 4.5.1

Running nonparametric bootstrap


Causal Mediation Analysis

Nonparametric Bootstrap Confidence Intervals with the BCa Method

               Estimate 95% CI Lower 95% CI Upper   p-value
ACME           0.014991     0.010165     0.019249 < 2.2e-16 ***
ADE            0.044602     0.026359     0.064367 < 2.2e-16 ***
Total Effect   0.059593     0.040616     0.080315 < 2.2e-16 ***
Prop. Mediated 0.251551     0.145553     0.354956 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Sample Size Used: 30000


Simulations: 50
```